



Strategic reward and altruistic punishment support cooperation in a public goods game experiment



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ABSTRACT

We compare the effects of and the motivations behind voluntary punishment and reward in a finitely repeated public goods game. Our experimental results show that (1) the level of cooperation is indistinguishable between the punishment and reward treatments when group membership does not change, but the reward treatment shows stronger endgame effects and that (2) punishment is much more effective than reward when group membership changes randomly between rounds. Strategic motivations are stronger in rewarding others, whereas more intrinsic motivations drive punishment behavior. We revisit the results of several previous experimental studies that are comparable to ours and show the overall consistency of our results with those in the prior experimental research.

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1. Introduction

Groups and societies use various forms of incentives to facilitate cooperation for mutual benefit. Incentives can be classified based on whether they are institutionalized or voluntary² and whether they are rewards or punishments. Among these, the effects of voluntary punishment have been extensively studied in the lab since the seminal works by Yamagishi (1986), Ostrom, Walker, and Gardner (1992), and Fehr and Gächter (2000).³ These studies and those that followed show that voluntary punishment is very effective for deterring free-riding and, thus, facilitating cooperation in diverse settings of public goods games (for example, Bochet, Page, & Putterman, 2006; Falk et al., 2005; Fehr & Gächter, 2002; Masclet, Noussair, Tucker, & Villaval, 2003; Page, Putterman, & Unel, 2005; Carpenter, 2007b; Carpenter, Bowles, Gintis, & Hwang, 2009).

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² Others use formal versus informal (Falk, Fehr, & Fischbacher, 2005, for example) or centralized versus decentralized (Gürerk, Irlenbusch, & Rockenbach 2009) instead of voluntary versus institutionalized.

³ The current paper concerns the effectiveness of and the motivations behind voluntary punishment and reward. However, we note that experimental studies of institutionalized or formal reward and punishment are also numerous. See, for example, Dickinson (1989), Dickinson and Isaac (1998), Tyran and Feld (2006), and many other works cited in Deci, Koestner, and Ryan (1999).

Because the punished and the punishers lose income when the option is exercised, researchers have questioned whether punishment is efficiency enhancing and evolutionarily viable (Bochet et al., 2006; Botelho, Harrison, Pinto, & Rutström, 2007). Gächter, Renner, and Sefton (2008) report that, with a sufficiently long time horizon, the availability of voluntary punishment raises the level of social welfare significantly compared to when the option is not available. In addition, subjects in public goods games choose the punishment option when they are allowed to vote or move (Ertan, Page, & Putterman, 2009; Güerker, Irlenbusch, & Rockenbach, 2006).⁴ Other issues include punishment directed towards high contributors (Herrmann, Thöni, & Gächter, 2008) and the inhibitory effect of retaliatory punishment when directed counter-punishment is possible (Denant-Boemont, Masclet, & Noussair, 2007; Nikiforakis, 2008). But in general, the findings show that voluntary punishment improves cooperation in public goods provision games.

Experimental studies of voluntary reward as a means to promote cooperation in N-person dilemmas are surprisingly rare given that our experience shows voluntary reward to be a plausible mechanism for solving the collective action problem. Being extra helpful to a neighbor with a hybrid car cannot be less plausible than snubbing a neighbor who drives a gas-guzzler. Voluntary reward is a form of gift-giving, an essential thread of social fabric across time and places (see works cited in Posner, 1981, in particular pp. 136–138, 159–161, 170–171). Though the evidence is not conclusive, a body of literature in psychology and education suggests that reward is a better incentive in changing attitude and behavior than punishment is (Banks, 2002; Becker, 1964; Kim, Shimojo, & Doherty, 2006; Larsen & Tentis, 2003; Skinner, 1971). In particular, studies of exchange relations in sociology suggest that reward may be a better means of achieving cooperation than punishment is (Molm, 1989, 1990; Molm, Quist, & Wiseley, 1993). Balliet, Mulder, and Van Lange (2011) conduct a meta-analysis to compare the effectiveness of punishment and reward, and examine the effects of various parameters on cooperation. This meta-analysis shows that punishment increases subjects' contribution in one-shot interactions whereas reward does not, but both have significant effects on the contribution in iterated games. However, only a small number of experimental studies directly compare the effectiveness of punishment and reward in repeated public goods games: Sefton, Shupp, and Walker (2007), Rand, Dreber, Ellingsen, Fudenberg, and Nowak (2009), and Sutter et al. (2010). These studies investigate the effects of punishment and reward in repeated public goods games, but report rather different results. Whereas Sefton et al. conclude that reward is less effective than punishment for achieving cooperation, Rand et al. argue that reward can be as effective as punishment is. Sutter et al. show that the relative effectiveness depends on the leverage factor of punishment. Comparing the results of these studies is difficult because they use different experimental protocols. Therefore there are still unresolved issues in comparing punishment and reward. For example, would Sefton et al.'s results still hold with a leverage factor of 1:3 instead of 1:1?⁵ Are the indefinite repetition and full reputation in Rand et al. necessary for reward to work? Would Sutter et al. still hold if subjects play the game in stranger condition? We will compare these results with ours in detail in the last section.

We answer these and related questions on the effects of reward and punishment. Our experiments also attempt to investigate the mechanisms by which reward and punishment work and to identify the motivation behind reward and punishment. This is done by comparing reward and punishment in both partner and stranger conditions.

Our findings on punishment are consistent with those reported by many other prior studies in that punishment is driven nonstrategically or by intrinsic motivation (Anderson & Putterman, 2006; Casari & Luini, 2012). Thus, punishment is a powerful facilitator of cooperation even in the stranger condition. This consistency of our results and the results of prior research conducted under similar protocols serves as a validity control as we examine the effects of reward and punishment. With a reward efficiency factor of three, we show that reward can be as effective as punishment without repeating the game indefinitely or providing full information to the rewarded on the identity of the reward giver. However, this is true only in the partner condition. In the stranger condition, reward becomes much less effective, whereas punishment remains as effective as in the partner condition. A large proportion of reward-giving appears to be strategically motivated.

This paper is organized as follows. Section 2 provides our experimental design. In Section 3 we discuss two possible motivations driving punishment and reward, and shows how our experimental design helps identifying individuals' motivation behind punishment and reward. Section 4 provides the results from our experiments and Section 5 compares some of the previous studies with ours.

2. Experimental design

Our experiment consisted of public goods games (voluntary contribution mechanism) with two different treatment conditions. We conducted 6 sessions, 3 sessions with reward treatment and 3 with punishment treatment. Subjects played the experimental game in groups of 3.⁶ In all 6 sessions, participants completed three consecutive phases (baseline phase, partner phase, and stranger phase), each of which was composed of 10 decision rounds. In other words, participants played 30 rounds of public goods games in total. In the first phase (baseline phase), subjects repeated a simple public goods game for 10 rounds

⁴ We do not mean here that the issue has been resolved one way or the other. Although Gächter et al. (2008) provide powerful evidence that, with a sufficiently long time horizon, the norm of voluntary punishment is efficient, several experimental studies have shown that subjects are reluctant to vote for the option (Botelho et al., 2007; Sutter, Haigner, & Kocher, 2010).

⁵ Changing the effectiveness ratios has been found to affect the effectiveness of punishment (Anderson & Putterman, 2006; Carpenter, 2007a; Egas & Riedl, 2008; Nikiforakis & Normann, 2008) and rewards (Vyrastekova & Soest, 2008). Most studies conclude that a higher leverage factor leads to higher cooperation.

⁶ The three benchmark studies (Rand et al., 2009; Sefton et al., 2007; Sutter et al., 2010) with which we later compare our results use groups of size 4. According to Balliet et al.'s meta-analysis (2011), group size does not moderate either the punishment-cooperation association or the reward-cooperation association.

Table 1

Experimental design: Three consecutive phases with two different treatment conditions.

Treatment	Baseline phase (Round 1–10, partner)	Partner phase (Round 11–20, partner)	Stranger phase (Round 21–30, stranger)	Number of sessions	Number of subjects/groups
Punishment	PGG	PGG/Punishment	PGG/Punishment	3	69/23
Reward	PGG	PGG/Reward	PGG/Reward	3	69/23

Note: PGG: Public goods game.

under the partner condition. In the second phase (partner phase), subjects played another set of 10 rounds, each of which was composed of two stages: a public goods game stage, and a punishment or reward stage. In the last phase (stranger phase), subjects repeated another 10 rounds of a public goods game followed by a punishment or reward stage under the stranger condition, where subjects were randomly assigned to new groups in each round so that subjects played the games with different partners each time (see Table 1).

The subjects' identification numbers changed between rounds. This prevented subjects from tracking other members' past behaviors and, as a result, limited reputation building.

In the public goods game stage, subjects were endowed with 20 tokens and asked to allocate them between a private account and a group account. Each token put into a subject's private account was worth 1 point to her, while each token in the public account was worth 0.5 points to every group member, including herself (i.e., $MPCR = 0.5$). Let x_i be subject i 's contribution to the group account and $g_i(x)$ be the earnings of subject i from the public goods game stage, where x is the vector of members' contributions. Then subject i 's earnings from the public goods game stage are determined by $g_i(x) = 20 - x_i + 0.5 \sum_j x_j$. After the subjects made decisions on x_i , they were informed of their own contribution, the total contribution to the public account in their group, and their own earnings from the public goods game stage.

In the partner phase and the stranger phase, another stage allowing punishment or reward followed the public goods game stage in each round. After deciding how much to contribute to the public account, each subject was informed of other members' contribution levels and their earnings from the public goods game stage with subjects' identification (A or B) randomly changing each round. Based upon this information, subjects were given an opportunity to reduce or increase other group members' points earned depending on whether they were in the punishment treatment or the reward treatment. Subjects were given an additional 10 tokens per round for this purpose. For each token used, three points were subtracted from or added to the target's earnings.

Six sessions were held in April and May 2009 at Kyungpook National University and Korea University.⁷ A total of 138 student subjects were recruited from the two universities by online advertisements on the universities' web bulletin boards. The experiment was programmed and conducted with z-Tree software (Fischbacher, 2007). Subjects were asked to arrive 10 min before the session started and were seated in the computer lab upon arrival. The session began by reading the instructions. The experimenter handed out the instructions to the participants and slowly read them aloud line by line, which took about 15 min (see Appendix A for the instructions). After reading the instructions, participants were asked to solve a series of quizzes to make sure that they clearly understood the instructions. Subjects were not allowed to talk or use any communication devices during the experiment.

Subjects were not informed that each phase would be followed by another phase. Subjects were told that the experiment would be conducted in an anonymous manner and their identities would not be revealed to others during or after the experiment.

Sessions lasted on average about 75 min, and subjects were paid in cash immediately after the experiment based on the points they earned during the session. One point was worth 30 KRW. The average total earnings were 34,168 KRW, with a maximum of 45,635 and a minimum of 26,825, including 5000 KRW participation fee.⁸

3. Strategic and intrinsic motivations behind punishment and reward

Many experimental studies have shown that people frequently punish norm-violators and reward norm-conformists even when punishment and reward are costly. Given the ample evidence provided by previous studies, it is clear that people tend to impose punishments and confer rewards. The problem that remains is to identify the motivations behind punishments and rewards and the conditions under which punishment or reward works better.

Two distinct motives may operate when subjects make decisions about whether or not to use the punishment and reward options. First, they may punish or reward others to induce higher contribution in future.⁹ In this case, individual i 's use of

⁷ For each treatment, two sessions were conducted in Kyungpook National University and one session in Korea University. The two treatments were balanced in terms of the composition of subjects from the two universities. No systematic difference was found between the two subject pools.

⁸ The average earnings total about \$30 when converted to U.S. dollars. The minimum hourly wage was 4110 KRW in 2009.

⁹ In the reward treatment, subject i might reward j because she believes that j will reward her later in return. This possibility is limited in our setting because individual identities do not persist round by round and reward is anonymously delivered to the target. However, expecting a counter-reward in the next round is not completely impossible in the partner phase if we take into account the small group size. We categorize the use of rewards in this motive as strategic use of reward.

punishment or reward depends on his or her expectation of future gain from punishing or rewarding others. We call this motive *strategic*. Secondly, intrinsic motive may drive the use of the punishment or reward option. Subjects punish others because they deviated from a social norm, or reward them because they obeyed the social norm. In this case, punishment and reward could be valued and exercised even when no future gains can be expected. Note that strategic consideration may play a role only in the partner condition whereas intrinsic motives may drive the use of punishment and reward in both the partner and stranger conditions. Therefore we assume the followings. (1) The punishments or rewards exercised in the stranger condition are from subjects' intrinsic motivation; (2) the punishments or rewards exercised in the partner condition are both from subject's strategic and intrinsic motivations. Therefore, comparing subjects' use of punishment (or reward) between the partner phase and the stranger phase, we can identify how much of the punishment (or reward) is from the strategic consideration or the intrinsic motivation.¹⁰

4. Experimental results

In this section, we provide detailed descriptions and statistical tests of the results. We will focus on contribution behavior and the effectiveness of punishment and reward in raising contribution levels in Section 4.1 and on rewarding and punishing behavior in Section 4.2.

4.1. Contribution to group account

Fig. 1 displays the time trends of the average contribution through the three phases for each treatment. The trends in the baseline phase are typical of the standard public goods provision experiments in that the levels of contribution start around 50% and then steadily decline over time. Introducing either reward or punishment in the partner phase immediately raises the contributions with apparent end-game effect only in the reward treatment. Reshuffling the groups every round in the stranger condition makes the punishment and reward less effective. The decrease of the contribution in the stranger phase is larger in the reward than in the punishment treatment. In the following we provide more rigorous and detailed analyses of the trends.

4.1.1. Result 1: Baseline phase versus partner phase. The level of contribution is higher in the partner phase than in the baseline phase in both the reward and punishment treatments

Fig. 1 and Table 2 show that in the punishment treatment the average contribution level increases from 8.17 in the baseline phase to 12.51 in the partner phase. In the reward treatment, the average contribution rises from 9.38 in the baseline phase to 12.31 in the partner phase. The difference is significant in both treatments in Wilcoxon matched-pairs signed-rank tests using group average contribution over 10 rounds as an observation ($N = 23$, $p < 0.01$).¹¹

4.1.2. Result 2: Reward versus punishment in partner phase. The partner phase contribution levels do not differ significantly between the reward and punishment treatments, but the reward treatment shows a larger end-game effect

In our experiment, both punishment and reward appear to be effective in raising the levels of contribution when groups remain intact. The average contribution levels in the partner phase are very similar between the two treatments (see Table 2 and Fig. 1). The difference is not significant ($p = 0.87$). However, the average contribution in the baseline phase was already somewhat higher in the reward (9.38) than in the punishment treatment (8.17). A difference-in-differences test shows that the increase of contribution in the reward treatment ($12.31 - 9.38 = 2.93$) and that in the punishment treatment ($12.51 - 8.17 = 4.34$) are not statistically different ($p = 0.2$).

One noticeable difference between the two treatments in the partner phase is the large decline of the contribution level by the end of the partner phase in the reward treatment, which is not observed in the punishment treatment (see Fig. 1). The average contribution level in the final round of the partner phase (9.09) is significantly lower than that in rounds 1 through 9 (12.67) in the reward treatment ($p < 0.01$). In the punishment treatment, on the other hand, contribution level in the final round (12.67) is actually higher than the average of the previous 9 rounds (12.49), though the difference is small and insignificant. However, the time trend in the partner phase of the reward treatment, we note, is not a steady decline as observed in many other public goods experiments but a large drop in the final round (see Fig. 1).

¹⁰ Efficiency concern might be one of the reasons for reward giving especially when the leverage factor is greater than one. But in the public goods games it is impossible to isolate the efficiency motivation from other motivations. Our design focuses on the strategic vs. intrinsic motivations and public good game is not suited for addressing the efficiency concern factor. Isolating efficiency concerns from other motivations has been done in other studies using modified dictator games (Andreoni & Miller, 2002; Charness & Rabin, 2002; Engelmann & Strobel, 2004).

¹¹ In this and all subsequent statistical tests, we used the Wilcoxon signed-rank test for comparisons of phases within treatments and the Wilcoxon rank-sum test for comparisons between treatments. Whenever possible, we use group average contribution over 10 rounds of a phase as the unit observation ($N = 23$). In statistical tests involving the stranger phase, however, the number of independent observations is 3 which is the number of sessions per treatment because of the random regrouping of subjects in each round. Therefore, in tests involving the stranger phases, we use the individual averages over 10 rounds as the unit observations. Using individual average as the unit observation violates the strict independence assumption. The test results, thus, need to be interpreted with qualifications.

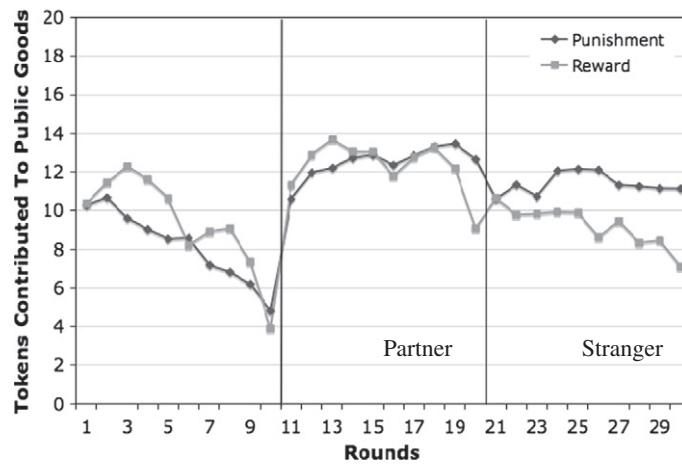


Fig. 1. The number of tokens allocated to the group account in the punishment and reward treatments.

Table 2

Punishment and reward levels in each phase.

	Punishment				Reward		
	Baseline phase	Partner phase	Stranger phase		Baseline phase	Partner phase	Stranger phase
Contribution	8.17	12.51	11.38	Contribution	9.38	12.31	9.21
Punishment	n.a.	0.98	1.50	Reward	n.a.	4.45	1.99
Punishment*	n.a.	0.96	1.48	Reward*	n.a.	4.74	2.04
Punishment**	n.a.	1.19	1.67	Reward**	n.a.	1.87	1.52

* Denotes the average excluding the last round of each phase.

** Means the punishment or reward level in the last round of each phase.

4.1.3. Result 3: Partner phase versus stranger phase. In both the reward and punishment treatments, the contribution level is significantly lower in the stranger phase than that in the partner phase. The decrease is larger in the reward than in the punishment treatment

In the reward treatment, the average contribution drops by more than three tokens between the partner and stranger phases (12.31 versus 9.21, $p < 0.01$).¹² In the punishment treatment, the average contribution drops by about one token in the stranger phase compared to the partner phase (12.51 versus 11.38, $p = 0.015$). The difference-in-difference test shows that the drop of contribution from partner phase to stranger phase is significantly larger in the reward treatment than in the punishment treatment ($p = 0.03$). It appears that both reward and punishment are more effective when used in repeated interactions among fixed partners, but the detrimental effect of random rematching is much stronger when groups use reward than when they use punishment.

As a result, the average contribution level in the stranger phase is significantly higher in the punishment treatment than in the reward treatment (11.38 versus 9.21, $p = 0.01$). We conclude that when groups are formed randomly in each round, voluntary punishment is more effective at sustaining cooperation than voluntary reward is.

4.2. Rewarding and punishing

4.2.1. Result 4: Subjects frequently use punishment and reward in both the reward and punishment treatments

Fig. 2 and Table 2 show that subjects frequently use punishment and reward in the partner phase. Subjects use, on average, 0.98 tokens per round for punishment and 4.45 tokens for reward in the partner phase. Punishment and reward levels during the stranger phase are also significantly greater than zero in both the punishment treatment and the reward treatment (1.45 tokens for punishment and 1.99 tokens for reward). Note that neither the cost of punishment nor the cost of reward can be compensated by future benefits under the stranger condition. The reward and punishment in the stranger condition, therefore, are unlikely to be driven by a strategic consideration. Instead, they reflect *intrinsic* motives for punishing free-riders and rewarding high contributors.

¹² Statistical tests related to Result 3 are done using individual averages ($N = 69$). See footnote 10 for details.

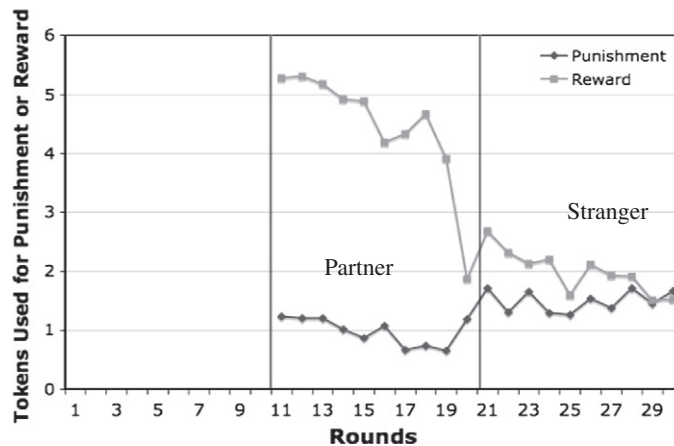


Fig. 2. The number of tokens used for punishment and reward.

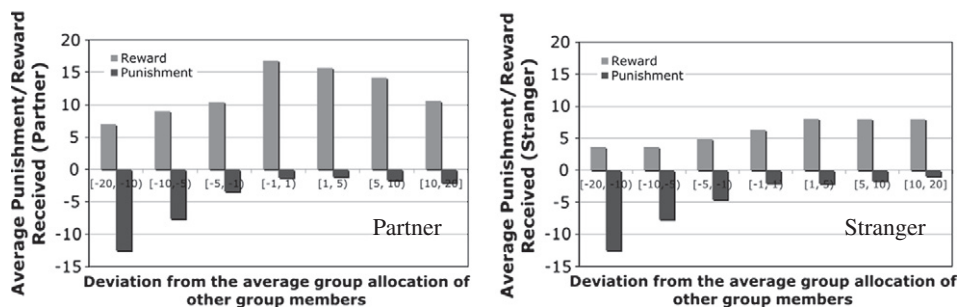


Fig. 3. Punishment and reward received in relation to deviation from other group members' average contribution.

4.2.2. Result 5: Subjects give larger rewards in the partner phase than they do in the stranger phase. Subjects' punishment behaviors do not significantly differ between the partner and stranger phases

In the partner phase, subjects spend on average 4.45 tokens per round out of a maximum of 10 to reward others, but they spend only 1.99 during the stranger phase (see Fig. 2 and Table 2). The difference is significant ($p < 0.01$, $N = 69$). To see if the lower reward level in the stranger phase is simply due to the lower amount of contributions we calculate the reward per unit of contribution (normalized reward) by dividing the average reward by the average contribution of the other group members. The normalized reward indicates how many tokens a subject gave per one token contribution by the other group members. During the partner phase it is 0.18, which is significantly larger than the corresponding value of 0.11 during the stranger phase ($p < 0.01$). We also compare the reward levels between the two phases, controlling for the target's contribution. We categorized all observations into seven groups according to their deviation from the average group allocation of other group members such that each category has approximately the same number of observations. We then calculated the average number of reward tokens received by a target in each group. Fig. 3 shows that, controlling for the level of contribution, the level of reward is much lower in the stranger phase than it is in the partner phase.

In the punishment treatment, the subjects on average spend 0.98 tokens to punish others in the partner phase. The size of punishment increases during the stranger phase such that the subjects on average spend 1.50 tokens to punish others ($p = 0.03$). To check if this difference reflects the somewhat lower levels of contribution during the stranger phase, as we did for the reward condition, we created a normalized punishment measure by dividing the level of punishment by the level of free-riding, where free-riding is defined as the number of tokens not contributed to the public account (i.e., 20 minus the contribution). The normalized punishment measure is 0.19 for the partner phase and 0.09 for the stranger phase. The difference is not significant ($p = 0.94$). In contrast to reward, the amount of punishment conditional on relative contribution does not differ between the partner and stranger phases. When we compare the punishment levels between the two phases at given levels of deviation from the average group allocation of other group members, we do not find that the punishment levels are different between the two phases (Fig. 3).

4.3. Statistical analysis of the determinants of reward and punishment

In a series of regression analyses, we study the determinants of punishment and reward in more detail. Table 3 presents the results of random-effects Tobit regressions, one for each treatment. The dependent variable is the number of tokens that

Table 3

Random-effects Tobit regressions of punishment and reward.

Dependent variable	Reward Coef. (Std. Err.)	Punishment Coef. (Std. Err.)
Constant	−3.89** (0.45)	−1.05 (0.70)
Partner phase dummy	3.20** (0.58)	0.07 (0.85)
Own contribution	0.15** (0.02)	0.14** (0.03)
Own contribution × Partner	−0.06* (0.03)	−0.03 (0.04)
Target's contribution	0.29** (0.02)	−0.32** (0.03)
Target's contribution × Partner	−0.04 (0.02)	−0.16** (0.04)
Third member contribution	−0.07** (0.02)	0.06* (0.02)
Third member contribution × Partner	−0.03 (0.02)	0.17** (0.04)
Reward/punishment received ($t - 1$)	0.00 (0.02)	0.06* (0.02)
Reward/punishment received ($t - 1$) × Partner	0.08** (0.02)	0.13** (0.03)
Round	−0.12* (0.05)	0.07 (0.07)
Round × Partner	−0.04 (0.07)	−0.21* (0.10)
Final round	0.07 (0.48)	−0.02 (0.54)
Final round × Partner	−2.29** (0.64)	0.94 (0.83)
The number of observation	2484	2484

* $p < 0.05$.** $p < 0.01$.

a player spends to reward (punish) a specific other player in the group in a round. In other words, the unit observation of the regression is the directed dyad in a round. The grouping variable for the estimation of the random-effect model is the subject identification number which is unique for each of the 69 individual subjects. We used the individuals, instead of the group, as the grouping variable because of the random re-matching of the subjects in each period of the stranger phase.

The variable *Own Contribution* codes the number of tokens that the sender of the reward (or punishment) contributed to the public account in that round. The variable *Target's Contribution* codes the contribution to the public account by the receiver of the reward (or punishment) in the dyad. The variable *Third Member's Contribution* codes the level of contribution by the third group member, who is not in the current dyad; including this variable is important because, as the results show, both the relative and the absolute contribution levels of the receiver of reward or punishment are important as a player decides how much to spend on rewarding or punishing the target.

The regression also includes the size of the reward or punishment that the rewarding or punishing player received in the immediately preceding round: *Reward/Punishment Received ($t - 1$)*, which codes the actual payoff change as a result of the total reward (punishment) received in the previous round or three times the sum of the rewards (punishments) that the other two members of the group sent in the previous round. To capture the time trend, *Round* is included. To capture the end-game effect, a final-round dummy variable is included. The regressions are fully interacted in the sense that all of the variables are interacted with *Partner Phase Dummy* and all of the interactive variables are included in the specification.

The regression results show that in both the partner and stranger phases: (1) the more one contributes for the public good, the more one rewards (punishes) others; (2) the higher the level of the target's contribution, the higher the level of reward (the lower the level of punishment); and (3) the higher the level of contribution by the third member of a group, the less a player rewards (the more a player punishes) the target. The second and third points suggest that both the absolute and relative contribution levels of a target matter when the giver of the reward or punishment decides how much to reward or punish the target.

The next two results are closely related to our main interest in this subsection, which is the difference between the partner and stranger phases.

4.3.1. Result 6. There is a large difference in the propensity to reward between the partner and the stranger phases, whereas no such difference is observed in the punishment treatment

This result confirms Result 5. The coefficient for the partner-phase dummy of the reward treatment is large (3.20) and significant, indicating that, controlling for all other factors, subjects spend about three more tokens to reward another member per round in the partner phase than they do in the stranger phase. However, no significant difference is found between the partner and stranger phases in the punishment treatment. The larger difference between the partner phase and the stranger phase in the reward treatment than in the punishment treatment suggests that reward is more strategically motivated than punishment is. Controlling for all other factors, the fixed grouping alone seems to generate a large incentive to reward others.

4.3.2. Result 7. The effects of reward and punishment in the previous round are significant in the partner phase

The reward received in the preceding period positively affects the size of the reward sent in the partner phase, but not in the stranger phase. This result suggests that reciprocity in reward-giving may affect the size of the reward. It is possible to infer from this result that part of the reason why one gives a reward is that one anticipates receiving a reward in return in the

Table 4

Effectiveness of reward versus punishment in finitely repeated public goods games.

	This paper ($L = 3$)	Sefton et al. (2007) ($L = 1$)	Sutter et al. (2010) ($L = 3$)	Sutter et al. (2010) ($L = 1$)
Group size	3	4	4	4
MPCR	0.5	0.5	0.4	0.4
Ratio of endowment for reward (punishment) to endowment for contribution	0.5	1	0.15	0.15
Effectiveness of reward and punishment	Both reward and punishment effective	Both reward and punishment effective	Both reward and punishment effective	Only punishment effective ^a
Contribution level with reward and punishment* (between-subject comparison)	$P = 12.51$ (62.6%), $R = 12.31$ (61.6%). Endowment = 20. Not significantly different	$P = 3.56$ (59.3%), $R = 3.57$ (59.3%). Endowment = 6. Data reported in Fig. 1	$P = 12.11$ (60.6%), $R = 11.20$ (56.0%). Endowment = 20. Not significantly different	$P = 8.79$ (44.0%), $R = 6.62$ (33.1%). Endowment = 20. Significance not reported
Reward and punishment as percentage of endowment	R: 45% P: 10%	R: 19% P: 25%	R: 44% P: 20%	R: 10% P: 15%

* P: Punishment; R: Reward.

^a One recent study by Nosenze and Sefton (2012) provides similar results to Sutter et al. that only punishment is effective in a partner treatment when the leverage factor is 1.

future. Though the experimental design does not allow full reputation building because the players are not assigned an identification number that remains constant for the duration of a phase, it was, we suspect, possible to build a reputation to some extent because of the small group size and the signaling effect of the contribution levels.

We also see a larger retaliatory component in the punishment during the partner phase, as indicated by the significant coefficient for *Punishment Received* ($t - 1$) $\times$ *Partner* variable. This suggests that the punishment during the partner phase is affected by the retaliatory motivations. Though the exact identity of the punisher cannot be known even in the partner phase, factors such as the small group size and perhaps an intuitive understanding that high contributors are the ones who punish probably made it sensible for the subjects to engage in some level of retaliatory punishment.

5. Discussion and conclusion

This section addresses two issues: (1) whether voluntary punishment or reward is more effective in facilitating and sustaining cooperation and (2) the sources of the difference, if any, between reward and punishment. Rand et al. (2009: p. 1272) argue that reward is at least as effective as punishment under the conditions of indefinite repetition and full reputation. Our experimental results indicate that, even in the more standard finitely repeated public goods game, reward is as effective as punishment, and this finding makes it necessary to revisit other experiments to develop a careful and nuanced generalization about the effectiveness of reward versus punishment.

Two other experimental studies (Sefton et al., 2007; Sutter et al., 2010) allow a comparison of the effectiveness of reward and punishment in a finitely repeated fixed matching condition. The experimental designs of the two studies are not identical. Sutter et al. use reward/punishment leverage of both 1-to-1 and 1-to-3, providing two conditions in which the comparison can be made.¹³ In all three experiments, including ours, we take the data from treatments in which subjects played 10 rounds of public goods games in fixed groupings and knew that the game consisted of a finite number of repetitions.

Table 4 summarizes the experimental designs and results of the three studies. All studies listed in Table 4 except for the 1:1 leverage condition in Sutter et al. find that voluntary reward and punishment are both effective in raising the contribution level compared to the baseline condition, and the difference in contribution levels between the reward and punishment conditions is not significant. Another common result among the three studies is a larger end-game effect in the reward condition, which is formally tested in the current paper and in Sefton et al. and visually verifiable in Table 1 of Sutter et al.'s work. As is well known, a large end game effect signifies that the incentives for cooperation are provided for strategic and self-interested reasons with uncertainties about other players' motivations (Kreps, Milgrom, Roberts, & Wilson, 1982). The evidence is consistent with a theory that subjects believe that others reward for strategic reasons and, thus, reward will decrease and become less effective as the game nears the end. However, the data in all three studies show that this is not the case with punishment. That is, many subjects believe that others would punish even at the end of the game, so it is better to contribute even in the final rounds. This belief seems to be justified given the substantial level of punishment used in the stranger conditions as well as in the final rounds of repeated games.

The stranger phases in our experiments allow a test of this conjecture that self-interest-based strategic consideration plays a larger role when using reward, whereas intrinsic motivations are more important when using voluntary punishment.

¹³ We consider only the exogenous treatments of Sutter et al., where the reward or punishment mechanisms are imposed on the subjects as opposed to the endogenous treatments in which subjects chose whether or not to use reward or punishment.

Indeed, as the results reported in Section 4.2 show, the contribution levels are significantly lower with reward than with punishment when groups are formed randomly in each round. In addition, the contribution levels in the stranger phase of the reward treatment are significantly lower than those in the partner phase, but the contribution levels of the stranger phase in the punishment treatment are not significantly different from those in the partner phase. Thus, punishment is a powerful mechanism for achieving cooperation regardless of whether the groups remain fixed, whereas reward is effective only when individuals interact repeatedly in fixed groups. Our data also suggest that, in fact, the intrinsic portions of reward and punishment observed at the end of the partner phases and during the stranger phases are not significantly different.

The experiments reported in this paper are designed to investigate the conditions under which voluntary reward and punishment options work to solve the collective action dilemma. We found that when individuals interact repeatedly in a partner matching both reward and punishment can sustain cooperation at levels much higher than when those options are not available. When individuals interact in a stranger matching, however, punishment is much more effective than reward. Therefore our results confirm Balliet et al.'s (2011) meta-analysis. As Balliet et al. suggest, what is less well known is how punishment and reward affect subjects' expectations of cooperation, intrinsic motivation or ethical concerns. Our experiments are designed to identify subjects' motivations behind the use of punishment and reward, and provide evidence that reward is used more strategically than punishment is. The patterns of contribution under the partner condition show a larger end-game effect in the reward treatment than in punishment treatment, and the rewards given to others are considerably smaller under the stranger condition than under the partner condition. Punishment, on the other hand, is more intrinsically motivated as suggested by the absence of an end-game effect and the similar patterns of use between the partner and stranger phases.

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Appendix A

A.1. Sample instructions (Baseline phase, English translation)

Welcome to the experiment. You are participating in an experiment about behavioral motivation and decision-making. This experiment is funded by the Korea Research Foundation.

Please read the following instructions carefully. You will receive a fixed amount of KRW 5000 for participating in this experiment and will be able to earn more money based on the decisions you make during the experiment. You will be paid in cash at the end of the experiment (other participants will not know your earnings).

From now on, you are not allowed to talk with each other or use any communication devices.

Your decisions will be processed anonymously, so there will be no chance for your identity to be revealed to other subjects in this room during the session.

If you have any questions during the instructions, please do not hesitate to ask.

You will be paid in cash immediately after this experiment based on the points you earn from the experiment.

1 point = 30 KRW.

(1) Group formation

Twenty-four students in this room are participating in the same experiment. When the session starts, you will become a member of a group with two other anonymous members. In other words, you will be a member of a group of 3. No identifying information will be revealed during the session, so you will not be able to know who your group members are.

This experiment is composed of 10 decision rounds. The group membership will not be changed for 10 rounds.

In each round, you will be asked to make a decision in the following situation.

(2) Experimental Procedure

In each round, you will be given 20 tokens. These are called endowment tokens. You will be asked to allocate these tokens between your private account and the group account. In each round, the following screen will appear (see Fig A1).

The top left corner of the screen informs you of the current round out of 10 rounds. On the top right corner of the screen, you see how many seconds you have to make the decision. You have 30 s to make a decision in each round. You have to decide how many of the 20 tokens you are going to contribute to the group account. Enter your decision in the blank box on the screen and click the red OK button to proceed. If you do not click the OK button within 30 s, a warning will appear (see Fig. A2). (Do not forget to click the red button. Your decision will not be finalized unless you click the red OK button.)

Round 1 of 10 남은 시간 [sec]: 26

당신에게 20개의 토큰이 주어졌습니다. 이 중 몇 개의 토큰을 공공계정에 기여하겠습니까?

OK

도움말
당신에게 주어진 20개의 토큰 중 몇 개를 공공계정에 기여할 것인지를 결정하는 단계입니다. 공공계정에 기여하고 남은 토큰은 여러분이 갖게 됩니다. 공공계정에 몇 개의 토큰을 기여할 것인지를 위 빈칸에 적어넣어주세요. 그리고 OK버튼을 눌러주세요.

Fig. A1. Screen for decision-making.

Round 1 of 10 남은 시간 [sec]: 0
시간이 경과되었습니다. 빨리 결정을 내려주세요.

당신에게 20개의 토큰이 주어졌습니다. 이 중 몇 개의 토큰을 공공계정에 기여하겠습니까?

OK

도움말
당신에게 주어진 20개의 토큰 중 몇 개를 공공계정에 기여할 것인지를 결정하는 단계입니다. 공공계정에 기여하고 남은 토큰은 여러분이 갖게 됩니다. 공공계정에 몇 개의 토큰을 기여할 것인지를 위 빈칸에 적어넣어주세요. 그리고 OK버튼을 눌러주세요.

Fig. A2. Warning sign.

You can revise your decision before you click the OK button. Once you click the OK button, your decision will be finalized, and you will not be able to revise your decision.

When all participants in this room finish making a decision, the following screen appears. (Before this screen, you may see a screen saying “Wait for a moment”.)

I will explain what each line on the screen means in a few minutes.

(3) Calculating your earnings in each round

• Earnings from your private account

I will explain how to calculate your earnings based on your decision. When you allocate a number of tokens to the public account, the remaining tokens will automatically be put into your private account.

The number of tokens in your private account = 20 - your contribution to the public account.

Example. If you contribute 10 tokens to the public account, you will have 10 tokens in your private account.

Example. If you contribute 20 tokens to the public account, you will have 0 tokens in your private account.

Example. If you contribute 0 tokens to the public account, you will have 20 tokens in your private account.

One token in your private account is worth 1 point. That is,

Your earnings from your private account = the number of tokens in your private account $\times$ 1 point.

When you have 10 tokens in your private account, you will get 10 points. 20 tokens in your private account give you 20 points, and 0 tokens in your private account give you 0 points. Remember, one point will be exchanged into 30 KRW.

• Earnings from the public account

When you contribute tokens to the public account, every member in your group will benefit. Every member (including yourself) will receive 0.5 times the number of tokens in the public account. For example, if there are 40 tokens in the public account, every group member will receive $0.5 \times 40 = 20$ points. That is, one token in the public account is worth 0.5 points to everyone in the group.

Example. Assume you do not contribute anything to the public account, while the total contribution from the other two members is 30. Everyone in the group will get 15 points from the public account.

Example. Assume you contribute 20 tokens to the public account and the sum of the other two members' contributions is 20. Then there will be 40 tokens in the public account. You will receive 20 points from the public account, and the other two members will also receive 20 points each.

Example. Assume you contribute 20 tokens to the public account and the other two members contribute nothing. Then there will be 20 tokens in the public account. You will receive 10 points from the public account, and the other two members will also receive 10 points each.

In sum, one token contributed to the public account is worth 0.5 points to every member of the group.

One token in your private account gives you 1 point. When you contribute one token to the public account, you will get 0.5 points instead of 1 point. However, 0.5 points will be given to the other members too. That is, one token in the public account raises 1.5 points in total for the group (0.5 times 3). You increase other members' earnings by contributing tokens to the public account. In short, one token contributed to the public account increases every member's earnings (including the contributor) by 0.5 point.

• Earnings in each round

Therefore, your earnings in each round will be determined in the following way.

Earnings from your private account = 20 - your contribution (1)

Earnings from the public account = $0.5 \times (\text{sum of all members' contributions})$ (2)

Your earnings = (1) + (2)

Fig. A3 shows the same information. The first two lines show your contribution to the public account and the total number of tokens in the public account.

The next two lines show your earnings from the private account and public account, respectively. The last line shows your total earnings from the round.

When you have reviewed the contents of the screen, click the CONTINUE button to proceed. The screen will remain for 15 s. After 15 s, you will move to the next round automatically. However, when you finish reviewing the screen showing your earnings, please click the button to facilitate the process.

A.2. Sample instructions (Punishment Phase, English translation)

Now we start another set of experiments. This experiment includes similar decision-making process to the one that you just finished, except that one additional decision making stage will be included in each round.

Round 1 of 10 남은 시간 [sec]: 15

이번에 당신이 기여한 토큰 수

당신이 속한 조의 공공계정에 모인 토큰의 총합

‘개인보유’로부터의 수익

‘공공계정’으로부터의 수익

이번에 당신의 수익

도움말
이번 기 당신이 기여한 토큰수와 공공계정에 모인 토큰의 총합, 그리고 그 결과 당신이 얻게 된 수입이 나타나 있습니다. 확인하시고 <계속> 버튼을 눌러주세요..

Fig. A3. Result of your decision.

Period 1 of 1 Remaining time [sec]: 0

앞의 표에는 당신이 속한 그룹에서 당신을 제외한 나머지 구성원들의 지난기 기여액이 (UEU) 있습니다.
이제 당신에게 20개의 토큰을 추가로 지급해드립니다. 이 토큰을 사용하여 다른 구성원들의 토큰 수를 지급할 수 있습니다.
당신이 토큰 한 개를 사용하면 상대방의 토큰 수가 1개 지급됩니다.
당 표의 빈칸에 상대방 토큰수를 지급하기 위하여 사용하고자 하는 토큰수의 수를 적어 넣으십시오. 만약 그대로 두어도 무방하다. 당신이 지급액에 사용한 토큰의 총합은 기록 남아서는 안됩니다.

	본인	구성원 A	구성원 B
공공계정에 의 기여액	xx	yy	zz
토큰 수익	000	000	000
지급액		0	0

Fig. A4. Screen for 2nd stage decision making.

This phase is also composed of ten rounds and each round has two decision-making stages. Same as before, group membership does not change once the group is formed.

1st Stage Decision: Same as before.

2nd Stage Decision:

- After every member decides how much to contribute to the public account, you will be informed of other members' contribution levels and their earnings from the first stage. Each member's contribution and earning will appear on the screen under the names of "Member A" and "Member B". Labeling of "Member A" or "Member B" will be randomly determined in each round and, thus, might change between rounds (see Fig. A4).

Period 1 of 1 Remaining time (sec) 52

당신의 이번기 기여액 _____

기여액 분할후 토큰 수 _____

다른 구성원의 토큰을 사용하여 위해 쓰인 토큰 수 (-) _____

추가 지급받은 토큰 (+) _____

소계 _____

타인에게 의한 당신의 토큰 삭감액 (-) _____

이번 회의 당신의 토큰 수합 _____

OK

Fig. A5. Result of your decision (2nd stage).

- You will be given 10 additional tokens for the 2nd stage decision making in each round.
- You will have an opportunity to reduce other members' earnings. You may reduce other members' earning points by using your tokens that you receive for this stage. For each token used, 3 points will be subtracted from the target's earnings. For example, you can reduce 6 points from member A's earning by using 2 tokens.
- The maximum number of tokens you can use for each target is 5. In other words, the maximum points that you can reduce from one member are 15.
- The unused tokens will be put into your private account (worth 1 point for each token).
- After deciding how many tokens to use to reduce other members' points, the information screen appears (See Fig. A5).
- Your earnings from each round is determined by the following formula;

$$1 \times (20 - \text{your contribution}) + 0.5 \times (\text{sum of all members' contribution}) + (10 - \text{the number of tokens you used for reducing other members' points}) - (\text{the reduction of your points due to other members' tokens assigned to reduce your earning}).$$
- The information provided in the information screen includes (1) your contribution and earning from 1st Stage; (2) the number of tokens you used to reduce other members' points; (3) the decrease in your points due to other members' decision; (4) your net earning in this round (For (3), only total amount of reduction will be shown, not how many tokens other members used, nor who used tokens for reducing your points.)
- Due to the payoff structure, your net earning may become negative. In this case, your earning in the round is set to be 0.

When you finish reviewing the screen or making a decision, please click <CONTINUE> button or <OK> button on the bottom right corner of the screen to proceed, which facilitate the process.

Note:

- The instruction for the reward treatment was written in a similar manner, except that we use the term “raise” instead of “reduce”.
- The instruction for the third phase is similar to the one for the second phase except that we said “In this phase your group membership changes every round” instead of saying “Same as before, group membership does not change once the group is formed”.

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